Temporal PIG: A Predict-Then-Verify Approach to Resisting the Noisy-TV Problem

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Abstract

Many curiosity-driven exploration methods in reinforcement learning are fundamentally open-loop: they reward predictions of learning without verifying that learning occurred. This makes them vulnerable to the Noisy-TV problem, where stochastic stimuli trap agents into endlessly seeking novelty from non-informative distractors. We propose a *predict-then-verify* principle: reward only the measured decrease in model uncertainty after each transition. We instantiate this as Temporal PIG (TPIG), a modification of Predicted Information Gain (PIG) that uses the actual change in KL-divergence as a temporal-difference reward. Evaluated on grid-world environments with varying distractor density, TPIG dramatically outperforms PIG on distraction rate, exploration speed, and model accuracy, demonstrating that closed-loop verification of learning is the key to resisting stochastic distractors.

Introduction

Reinforcement learning (RL) agents learn by taking actions in an environment and receiving rewards that guide future behavior (Sutton and Barto 2018; Kaelbling, Littman, and Moore 1996). When environmental rewards are sparse or absent, agents must generate their own *intrinsic* rewards to drive exploration (Aubret, Matignon, and Hassas 2019; Schmidhuber 2010; Oudeyer, Kaplan, and Hafner 2007). Common intrinsic reward methods include count-based novelty (Tang et al. 2016; Bellemare et al. 2016), prediction-error curiosity (Pathak et al. 2017; Houthooft et al. 2016), and information-theoretic approaches (Little and Sommer 2013).

A fundamental weakness of many intrinsic motivation methods is the *Noisy-TV problem* (Burda et al. 2019a). Consider an environment containing a television with a remote control. Each button press displays a random, never-before-seen channel. Novelty-seeking agents become trapped, perpetually pressing the remote because each press yields a "new" state, even though no useful information about the environment is being learned.

Predicted Information Gain (PIG) (Little and Sommer 2013) selects actions by estimating, via KL-divergence, how much each action would improve the agent's transition model. Although PIG is more principled than raw novelty-seeking, it still falls prey to the Noisy-TV: the stochastic transitions from button presses yield high predicted divergence, luring the agent back to the distractor.

The core issue is that PIG *predicts* information gain but never *confirms* that learning actually occurred. We propose a **predict-then-verify** approach: compute the predicted information gain both before and after incorporating a new transition, and use the *actual decrease* as a TD reward. We instantiate this as Temporal PIG (TPIG) and show that it resists the Noisy TV because stochastic distractors produce large predictions but negligible measured improvement.

Related Work

Intrinsic Motivation for Exploration

Intrinsic motivation methods address the sparse-reward problem by letting agents generate their own learning signals (Aubret, Matignon, and Hassas 2019; Schmidhuber 2010). This idea has been realized in several families of algorithms.

Count-based methods reward visiting novel states by tracking visitation counts. Bellemare et al. (2016) introduced pseudo-counts to unify count-based exploration with intrinsic motivation in high-dimensional spaces, and Tang et al. (2016) extended this approach using hash functions. However, count-based methods are directly vulnerable to the Noisy-TV problem: every channel change registers as a novel state, producing unbounded reward from a single distractor.

Prediction-error methods use the agent's inability to predict outcomes as a reward signal. Pathak et al. (2017) proposed the Intrinsic Curiosity Module (ICM), which trains a forward dynamics model and uses its prediction error as intrinsic reward. Houthooft et al. (2016) proposed VIME, which maximizes information gain about environment dynamics using Bayesian neural networks. Burda et al. (2019b) introduced Random Network Distillation (RND), which measures novelty as the prediction error of a randomly initialized fixed network. While these approaches are more robust than raw count-based methods, Burda et al. (2019a) demonstrated in a large-scale study that prediction-error agents can still be attracted to inherently unpredictable transitions such as those produced by a Noisy TV.

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Information-theoretic methods, including PIG (Little and Sommer 2013), estimate the expected change in the agent's world model using information-theoretic quantities such as KL-divergence. Little and Sommer (2013) showed that PIG-based exploration learns environment structure faster than reward-free baselines and transfers well to downstream goal-directed tasks. Ménard et al. (2020) studied active learning for pure exploration in RL, providing theoretical analysis of sample-efficient exploration strategies.

The Noisy-TV Problem

The Noisy-TV problem was identified as a challenge for curiosity-driven agents by Burda et al. (2019a) and has since become a standard test of robustness for intrinsic motivation methods (Kovac, Laversanne-Finot, and Oudeyer 2020). Kovac et al. (2020) proposed goal-sampling strategies based on learning progress to mitigate sensitivity to noise. Ladosz et al. (2022) survey the broader exploration literature and note that most intrinsic motivation methods remain susceptible to stochastic distractors, identifying this as an open problem.

TPIG can also be viewed as an instantiation of *learning progress* (Oudeyer, Kaplan, and Hafner 2007), which rewards reductions in prediction error over time. TPIG operates directly on PIG's KL-divergence formulation without auxiliary networks, making it a minimal modification. The same predict-then-verify principle could be applied with other divergence measures such as MSE.

Background

Temporal Difference Learning

In temporal difference (TD) learning (Sutton and Barto 2018), the agent maintains Q-values $Q(s, a)$ for each state-action pair (Watkins and Dayan 1992) and updates them after each transition (s, a, s') using:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right], \tag{1}$$

where r is the reward, α is the learning rate, and γ is the discount factor (Sutton and Barto 2018).

Predicted Information Gain (PIG)

PIG uses the KL-divergence (Kullback and Leibler 1951) to estimate how much an agent's transition model would change if it took a particular action. Let $\hat{\Theta}_{sa\cdot}$ denote the agent's estimated transition probabilities from state s under action a, and let $\hat{\Theta}^{s \to s^*}_{sa\cdot}$ denote the updated estimate after hypothetically observing a transition to s^*. The predicted information gain is:

$$\mathrm{PIG}(s,a) := \sum_{s^*} \hat{\Theta}_{sas^*} \, D_{\mathrm{KL}} \left(\hat{\Theta}^{s \to s^*}_{sa\cdot} \parallel \hat{\Theta}_{sa\cdot} \right), \tag{2}$$

where D_{KL} is the KL-divergence. PIG selects the action a that maximizes this quantity.

The Noisy-TV Problem

When the environment contains a Noisy TV, the stochastic channel changes produce transitions to many distinct states, each with high predicted information gain. PIG's action selection thus repeatedly favors the TV because the weighted average over all possible channel outcomes remains large, even though the transitions carry no useful structural information about the environment.

The Predict-Then-Verify Principle

TPIG modifies PIG by measuring the *actual* information gained after each transition rather than relying on predictions alone. After observing transition (s, a, s'), TPIG computes the predicted information gain both before and after updating the internal model:

$$r_{\mathrm{TPIG}}(s,a,s') := D_{\mathrm{KL}}\big(\hat{\Theta}^{s \to s',\, t-1}_{sa\cdot} \parallel \hat{\Theta}^{t-1}_{sa\cdot}\big) - D_{\mathrm{KL}}\big(\hat{\Theta}^{s \to s',\, t}_{sa\cdot} \parallel \hat{\Theta}^{t}_{sa\cdot}\big), \tag{3}$$

where superscripts $t-1$ and t denote the model before and after incorporating the observed transition. The first term is the predicted information gain *before* the update; the second is the predicted gain *after*. If genuine learning occurred, the remaining information decreases and the reward is positive.

Note that both KL-divergence evaluations reuse the same computation that PIG already performs, so TPIG adds only a single additional divergence calculation per time step. This reward is used to update Q-values via standard TD learning (1), with $\alpha = 0.9$ and $\gamma = 0.9$. Q-values are initialized to 1, providing an optimistic prior that encourages exploration of all actions before learned rewards dominate. TPIG selects the action with the highest Q-value:

$$\mathrm{TPIG}(s) = \arg\max_{a} Q(s,a). \tag{4}$$

The Closed-Minded Model Problem

A subtle issue arises in the PIG algorithm when a state-action pair has been observed only once. After a single observation of transition (s, a, s'), the agent records the probability of that transition as 100%. On subsequent visits to state s, PIG predicts that taking action a will deterministically lead to s', making the predicted and observed distributions identical. The KL-divergence returns 0, and PIG considers the action fully learned, even if the true transition is stochastic with additional possible outcomes.

This "closed-minded" behavior causes PIG to neglect areas it has visited only once, focusing instead on regions where it has already observed variance. Both PIG and TPIG address this by initializing the total observed transition count for each state-action pair to 1 rather than 0. This is equivalent to placing a uniform Dirichlet prior on the transition probabilities, ensuring that the estimated and predicted distributions are never identical, so the KL-divergence always indicates some remaining information to be gained. As observations accumulate, the prior is overwhelmed by data and the model converges to the true distribution.

Handling Infinities

When $\hat{\Theta}_{sa\cdot}$ is empty (no transitions observed), the KL-divergence returns infinity. These values are replaced with

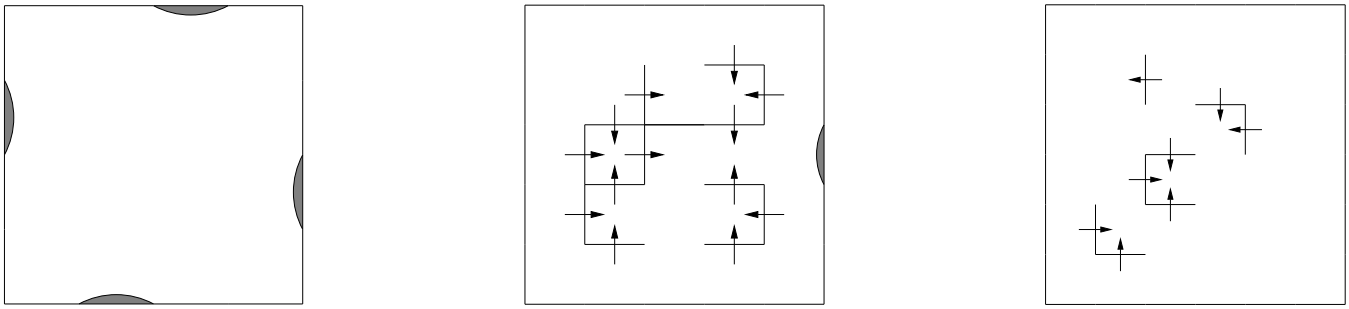


Figure 1: Grid-world simulations. Semi-circles are Noisy TVs; arrows indicate one-way walls. Left: 4-TV (4×4). Center: 1-TV (5×5). Right: No-TV (6×6).

0 in the reward calculation. Negative rewards (which can occur when discovering a new state temporarily increases uncertainty) are clamped to 0 to avoid penalizing exploration. Similarly, any Q-values that are infinite are normalized to 0 before the TD update is applied.

Why TPIG Resists the Noisy TV

When the agent presses the TV remote, it transitions to a "new" channel state. PIG predicts high information gain because the model would change substantially. However, after the update, TPIG recomputes the predicted gain and finds it is still high because the TV states are so numerous that each individual observation barely reduces the remaining uncertainty. The resulting reward is near zero, and TPIG's Q-values for TV-related actions decay, redirecting the agent to parts of the environment where observations actually improve the model. More precisely, for a purely stochastic transition with k equally likely outcomes, the n-th observation reduces the predicted information gain by only $O(1/n)$, so the TPIG reward vanishes as observations accumulate. Structured transitions, by contrast, produce larger reductions that persist until the model converges. In summary, PIG takes the action it *predicts* will teach it the most. TPIG takes the action *experienced* to yield the most actual learning, a crucial distinction that makes the predict-then-verify approach robust to stochastic distractors.

Epsilon-Greedy Variants

We also evaluate ϵ-greedy versions of both algorithms: EPIG and ETPIG. With probability ϵ (set to 10% unless otherwise noted), the agent selects a random action instead of the greedy choice.

Experimental Setup

Grid-World Simulations

We test on three grid-world environments of increasing size (Figure 1):

- **4-TV** (4×4): Multiple Noisy TVs.
- **1-TV** (5×5): A single Noisy TV.
- **No-TV** (6×6): No distractors (control).

In each environment, actions move the agent in one of four cardinal directions. Movement succeeds with 90% probability; with 10% probability the agent remains in place. Walking into a wall always leaves the agent stationary. Activating a Noisy TV increments an unbounded noise variable in the state, so each activation produces a unique state, simulating a TV with an inexhaustible supply of channels. The noise variable does not affect actual transition dynamics: all TV states share the same transition probabilities regardless of channel. The noise is therefore a pure distractor that inflates the apparent state space without adding any learnable structure.

All agents start in position 5. Each episode runs for 1000 time steps with no goal state. Every policy is run for 100 independent trials per simulation. The source code and simulation data will be made publicly available upon publication.

Metrics

- **Distraction rate**: Cumulative Noisy-TV activations.
- **Learning rate**: Number of undiscovered state transitions remaining.
- **Model accuracy**: KL-divergence between the agent's model and the true transition probabilities (excluding undiscovered transitions, which would contribute infinity).

Policies Compared

Five policies are evaluated: Random Walk (uniform random baseline), PIG, EPIG (ϵ-greedy PIG), TPIG, and ETPIG (ϵ-greedy TPIG).

Results

Distraction Rate

Figures 2 and 3 show cumulative distractions in the 4-TV and 1-TV environments. The PIG algorithm is heavily distracted by the Noisy TVs, with its distraction count growing steadily throughout the simulation. The Random Walk baseline, which selects actions uniformly at random, is distracted the least simply because it has no bias toward any particular action.

TPIG is initially distracted at a rate comparable to PIG, as it has not yet accumulated enough experience to distinguish

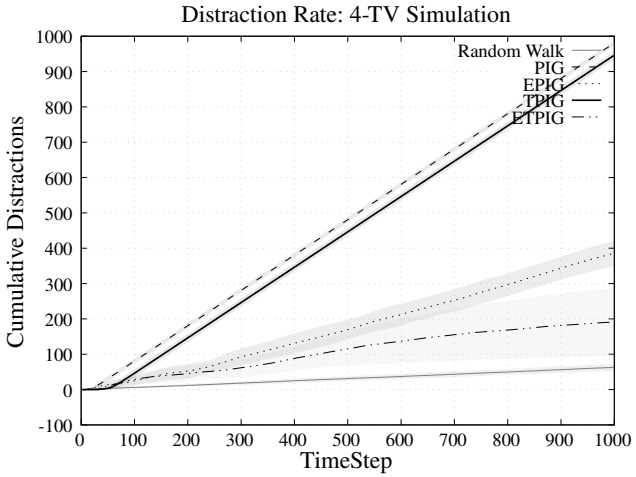


Figure 2: Distraction rate: 4-TV simulation. Shaded regions show ± 1 standard deviation over 100 trials.

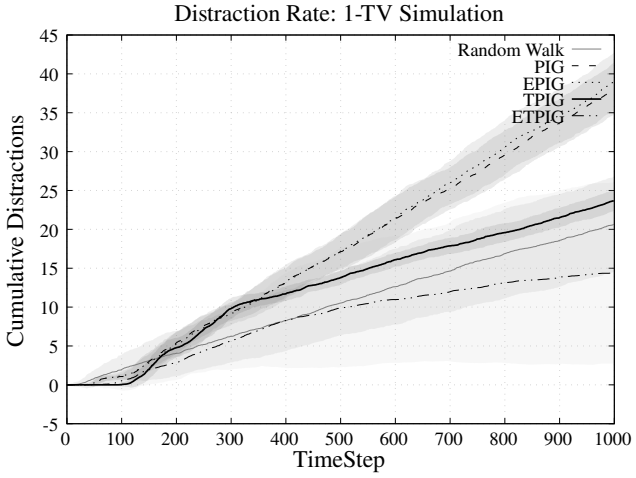


Figure 3: Distraction rate: 1-TV simulation.

informative transitions from noise. However, after a brief learning period, TPIG sharply reduces its rate of distraction. In the 1-TV environment (Figure 3), TPIG's distraction rate approaches that of Random Walk, indicating that it has effectively learned to avoid the TV. In the 4-TV environment (Figure 2), the improvement is less dramatic due to the higher density of distractors, but TPIG still substantially outperforms PIG. By the final time step, TPIG reduces cumulative distractions by 65% relative to PIG in the 4-TV environment and by 88% in the 1-TV environment.

The ϵ-greedy variants EPIG and ETPIG both show reduced distraction compared to their greedy counterparts, as the random action selection helps escape distraction loops. In the No-TV control environment (not shown), all algorithms produce identical flat lines at zero since there are no distractors.

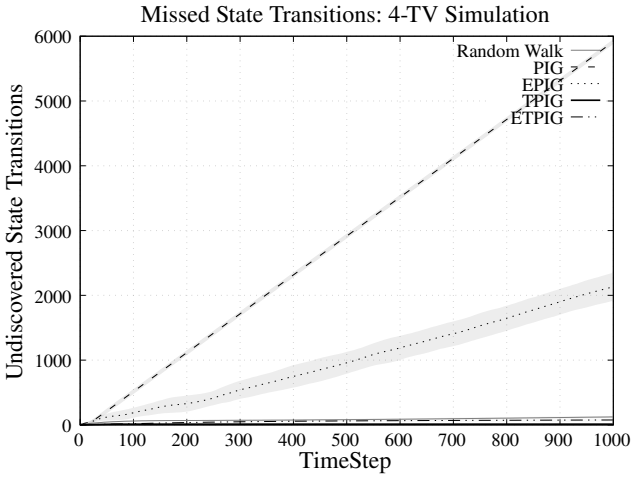


Figure 4: Missed state transitions: 4-TV simulation.

Learning Rate

Figures 4, 5, and 6 show the number of undiscovered state transitions over time. PIG initially explores faster than Random Walk in the 1-TV (Figure 5) and No-TV (Figure 6) environments, owing to its directed exploration strategy. However, in the 4-TV environment (Figure 4), PIG's initial advantage disappears as it becomes trapped by the Noisy TVs, ultimately performing worse than Random Walk.

TPIG starts slower than both PIG and Random Walk in all three environments. This is expected: TPIG must first accumulate enough transitions to build meaningful Q-values before its directed exploration becomes effective. After this initial learning phase, TPIG consistently surpasses both PIG and Random Walk, discovering new transitions at a faster rate. This crossover occurs earlier in smaller environments and later in larger ones.

The ϵ-greedy variants show no significant impact on learning rate, indicating that at $\epsilon = 10\%$, the random perturbation is too small to meaningfully affect the rate of new state discovery.

Model Accuracy

Figures 7, 8, and 9 show the KL-divergence between each agent's estimated model and the true transition model. Interpreting this metric requires care: because undiscovered transitions would contribute infinity to the KL-divergence, they are excluded from the summation. Consequently, an agent that has discovered more states may report a higher KL-divergence simply because more terms are included. Model accuracy must therefore be evaluated jointly with the learning rate results.

In the 4-TV environment (Figure 7), PIG shows worse KL-divergence than Random Walk and has discovered fewer states by the end of the simulation, placing it clearly behind the baseline. In the 1-TV environment (Figure 8), the comparison is closer: PIG has a slightly worse KL-divergence but has discovered more states, making the two roughly equivalent.

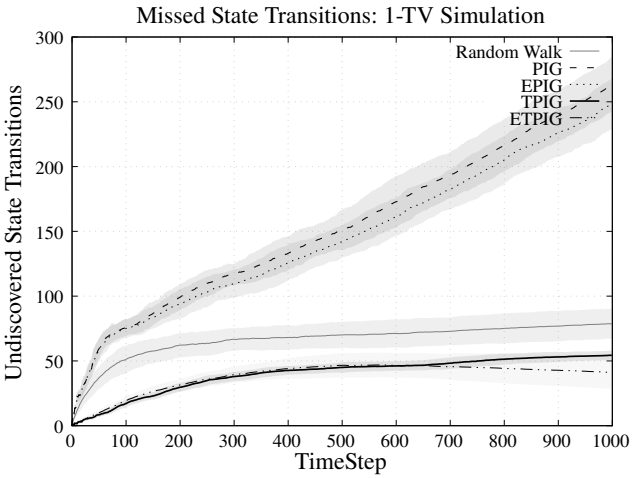


Figure 5: Missed state transitions: 1-TV simulation.

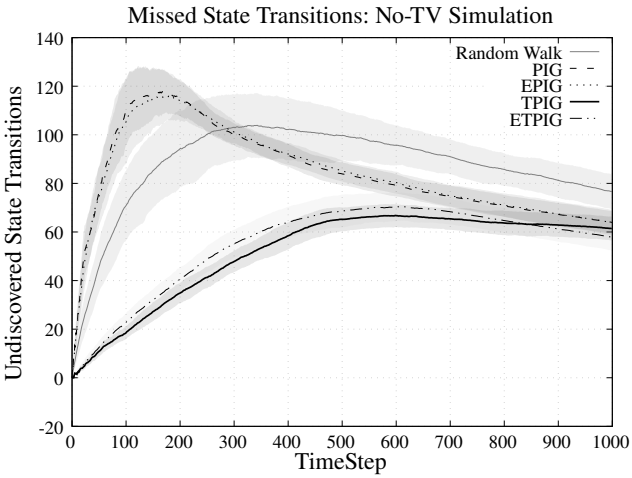


Figure 6: Missed state transitions: No-TV simulation.

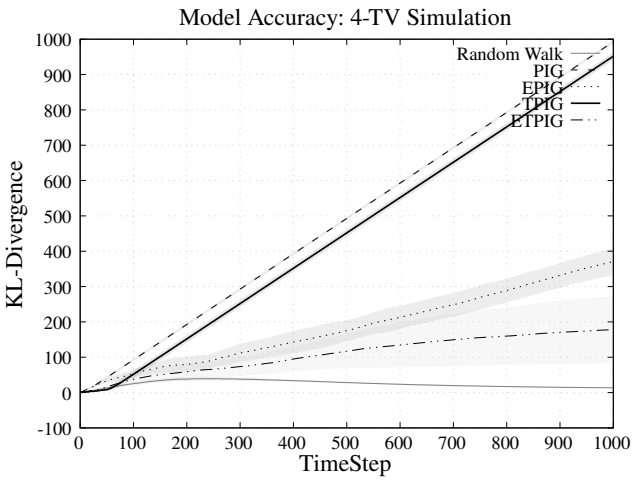


Figure 7: Model accuracy (KL-divergence): 4-TV simulation.

TPIG achieves lower KL-divergence than both PIG and Random Walk at all time steps across all three environments, while simultaneously having discovered more transitions. This combination indicates that TPIG both explores faster and learns the transition structure more accurately. The advantage is especially pronounced in the 4-TV environment, where PIG wastes substantial time on non-informative TV interactions.

In the No-TV control (Figure 9), where there are no distractors, PIG outperforms Random Walk and TPIG performs best. This confirms that TPIG's advantage is not merely an artifact of avoiding distractors; it produces genuinely better exploration even in clean environments. The ϵ-greedy variants perform comparably to their base algorithms, except in the No-TV case where random perturbation slightly hurts performance by diverting the agent from its optimal exploration path. Across all three metrics, TPIG's closed-loop verification consistently outperforms PIG's open-loop predictions in both noisy and clean environments.

Effect of Epsilon

We tested ϵ values in increments of 20% for both EPIG and ETPIG. In the 4-TV simulation, higher ϵ reduces distractions for both algorithms. For ETPIG, all ϵ values converge to similar learning rates by time step 250. In the 1-TV and No-TV simulations, lower ϵ values yield better long-term performance for ETPIG, suggesting that TPIG's learned avoidance makes heavy randomization counterproductive. Pure TPIG ($\epsilon = 0$) exhibits early artifacts smoothed by even small ϵ values, suggesting that modest randomization stabilizes TPIG's learning dynamics. For PIG-based methods, higher ϵ is consistently helpful since PIG lacks any mechanism to learn from its mistakes and requires external randomization to escape distraction loops. This asymmetry further confirms that the predict-then-verify mechanism provides its own form of learned exploration control.

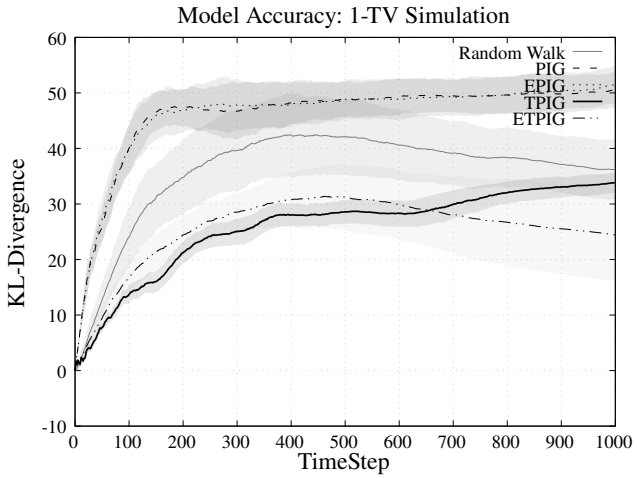


Figure 8: Model accuracy (KL-divergence): 1-TV simulation.

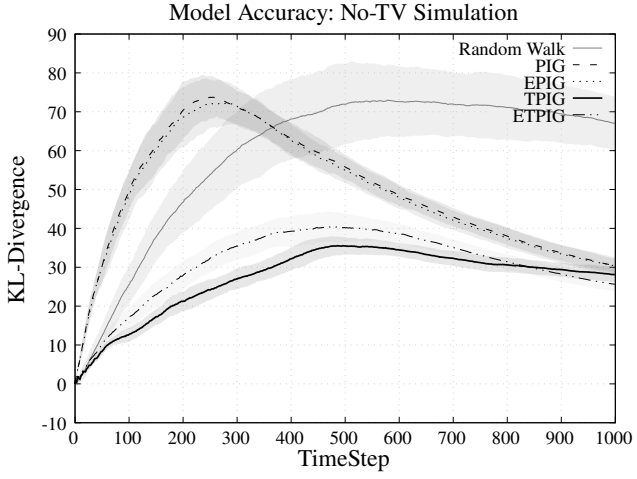


Figure 9: Model accuracy (KL-divergence): No-TV simulation.

Discussion

By conditioning rewards on the measured decrease in model uncertainty, the predict-then-verify approach transforms exploration into a closed-loop feedback system. The Noisy TV generates large predictions but negligible actual learning, so Q-values for TV-related actions naturally decay. Because the reward converges to zero as the model approaches the true distribution, TPIG inherits TD convergence guarantees (Sutton and Barto 2018), unlike open-loop curiosity methods that can be permanently misled by stochastic stimuli.

TPIG's Learning Dynamics

TPIG exhibits a characteristic two-phase behavior. In the first phase, TPIG explores broadly and somewhat inefficiently while accumulating the transition data needed to compute meaningful rewards. During this phase, PIG may outperform TPIG on learning rate because PIG immediately pursues the highest-predicted-gain actions. In the second phase, TPIG's Q-values have stabilized enough to reliably distinguish informative actions from noisy ones, and TPIG surpasses PIG on all metrics. This pattern is consistent across all 100 trials, as the narrow error bands in the figures confirm.

The duration of this initial learning phase scales with environment size. In the 4×4 4-TV environment, the crossover occurs around time step 200; in the 5×5 1-TV environment, around time step 300; and in the 6×6 No-TV environment, around time step 400. This approximately linear scaling suggests that TPIG's startup cost is predictable and proportional to the number of state-action pairs, representing 20–40% of the episode length across the environments tested. The learning rate $\alpha = 0.9$ and discount factor $\gamma = 0.9$ were chosen to favor rapid adaptation; preliminary experiments suggest the results are robust for all $\gamma \geq 0.5$, with low α providing a conservative mode that prioritizes distraction avoidance over exploration speed.

The Role of Epsilon

The interaction between ϵ-greedy exploration and TPIG reveals an important property. In environments with many distractors (4-TV), higher values of ϵ help both EPIG and ETPIG escape distraction loops. However, in environments with few or no distractors (1-TV, No-TV), lower values of ϵ yield better long-term performance for ETPIG. This asymmetry arises because TPIG already learns to avoid distractors on its own; adding random exploration on top of an already-effective avoidance strategy simply introduces noise that slows convergence.

For PIG-based methods (EPIG), higher ϵ is consistently helpful in environments with distractors, since PIG lacks any mechanism to learn from its mistakes and requires external randomization to escape.

Comparison with Other Approaches

Unlike RND (Burda et al. 2019b) or ICM (Pathak et al. 2017), TPIG requires no auxiliary networks, no explicit distractor detection, and no prior knowledge of which states are noisy. Its noise resistance emerges naturally from the closed-loop

reward structure and also improves exploration in clean environments, as the No-TV results demonstrate. Because the predict-then-verify mechanism is independent of the specific divergence measure, practitioners could substitute MSE or other metrics for KL-divergence with comparable results.

Limitations

The current study has several limitations. First, the grid-world environments are tabular. Scaling to continuous state spaces would require function approximation for the transition model and Q-values. Second, our unbounded noise model (infinitely many unique TV states) represents a worst case for novelty-based methods; real-world distractors with finite state spaces would be easier to overcome. Third, we do not combine TPIG's intrinsic reward with extrinsic rewards for goal-directed tasks. However, the $\pm 1\sigma$ bands in all figures confirm low inter-trial variance, indicating that the results reported here are statistically robust despite these limitations.

Conclusion

We introduced the predict-then-verify principle for curiosity-driven exploration, instantiated as TPIG. Across three grid-world environments with zero to four distractors, TPIG outperformed PIG on all metrics: fewer distractions, faster exploration, and more accurate models. Because the principle depends on measured improvement rather than a specific divergence measure, it could equally be instantiated with MSE or other metrics, and integrated into deep RL curiosity methods such as ICM (Pathak et al. 2017) or RND (Burda et al. 2019b). The choice of divergence measure becomes a tunable design parameter rather than a fundamental algorithmic decision. Future work includes scaling to continuous environments, testing with MSE and other divergence measures, and combining intrinsic and extrinsic rewards.

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